# Computer Vision Concepts

Computer vision is an area of artificial intelligence that helps computers interpret and work with visual information such as photographs, video frames, diagrams, and scanned documents. A computer-vision system may identify objects, detect faces, read text, measure movement, or separate an image into meaningful regions. For example, a navigation application can use cameras to recognize road signs, while a manufacturing system can inspect products for visible defects. Modern computer vision commonly uses deep-learning models trained on many labeled images. The model learns visual patterns such as edges, shapes, textures, and object parts. Good performance depends on diverse training data, appropriate evaluation, and careful handling of lighting, camera angles, and image quality. Because visual systems can affect privacy and safety, developers should also consider consent, accuracy across groups, and the consequences of incorrect predictions.